Unexpected and expected shocks under time to produce

October 2026

Household

$$\max \sum_t \beta^t \, \Xi_t \Big[\frac{C_t^{1-\gamma}}{1-\gamma} - \chi \frac{L_t^{1+\eta}}{1+\eta} \Big] \quad \text{s.t.} \quad P_t C_t + B_t = W_t L_t + (1 + i_{t-1}) B_{t-1} + \Pi_t$$

$$\Xi_t \frac{C_t^{-\gamma}}{P_t} = \beta (1 + i_t) \, \Xi_{t+1} \frac{C_{t+1}^{-\gamma}}{P_{t+1}}, \qquad \frac{W_t}{P_t} = \chi L_t^{\eta} C_t^{\gamma}$$

Ξ_t: demand shock. $\gamma = 2, \ \eta = 2, \ \chi = 1, \ \rho = 1\%.$

Final good

$$Y_t = \Big(\sum_i y_{i,t}^{\frac{\sigma-1}{\sigma}}\Big)^{\frac{\sigma}{\sigma-1}}, \qquad y_{i,t} = \Big(\frac{p_{i,t}}{P_t}\Big)^{-\sigma} Y_t, \qquad P_t = \Big(\sum_i p_{i,t}^{1-\sigma}\Big)^{\frac{1}{1-\sigma}}$$

Competitive. $\sigma = 5$.

Intermediate producers

Labour hired at t delivers Δ periods later:

$$y_{i,t}(\Delta) = \kappa^{\Delta}\,(\Delta + N_i)\,A_t\,l_{i,t}$$

Competitive, price-taking:

$$\max_{\Delta,\, l_{i,t}} \; p_{i,t+\Delta}\Big(\prod_{j=t}^{t+\Delta-1} \frac{1}{1+i_j}\Big)\, y_{i,t}(\Delta) - W_t\, l_{i,t}$$

A_t: TFP shock, dated at hiring.

Delay and price

Delay

$$\tau_i = \arg\max_{\Delta} \; (\beta\kappa)^{\Delta}(\Delta + N_i)$$

$$\Delta_i^* = -\frac{1}{\log(\beta\kappa)} - N_i$$

Rate enters unit cost: **cost channel**.

Price (zero profit)

$$p_{i,t} = \frac{W_{t-\tau_i} \prod_{j=t-\tau_i}^{t-1}(1+i_j)}{\zeta_i \, A_{t-\tau_i}}$$

$$\zeta_i = \kappa^{\tau_i}(\tau_i + N_i)$$

Ideal price index

Fixed wage, $W_t = W$:

$$P_t = \Big(\sum_i p_{i,t}^{1-\sigma}\Big)^{\frac{1}{1-\sigma}} = \frac{W}{\tilde{A}_t}, \qquad \tilde{A}_t = \left(\sum_i \Big[\zeta_i\, A_{t-\tau_i} \prod_{j=t-\tau_i}^{t-1} \frac{1}{1+i_j}\Big]^{\sigma-1}\right)^{\frac{1}{\sigma-1}}$$

Steady state, $i = \rho$:

$$\tilde{A} = \Big(\sum_i (\beta^{\tau_i}\zeta_i)^{\sigma-1}\Big)^{\frac{1}{\sigma-1}}$$

P_t moves only with rates and TFP over the last $\max \tau_i$ periods: $\pi_t = -\Delta \log \tilde{A}_t$.

Market clearing

$$C_t = Y_t, \qquad L_t = \sum_i l_{i,t} = \sum_i \frac{y_{i,t+\tau_i}}{\zeta_i A_t}, \qquad B_t = 0$$

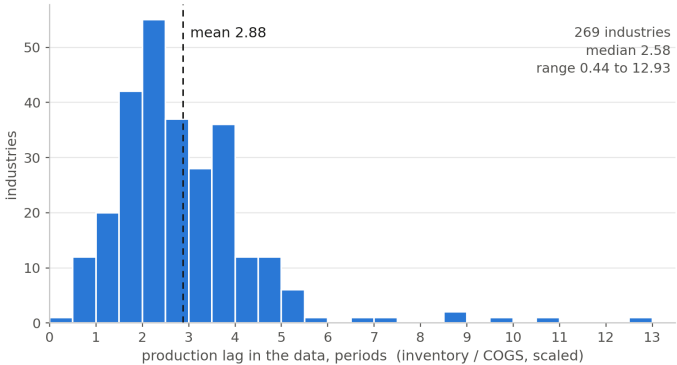
Employment at t produces deliveries at $t + \tau_i$.

Labour today, output today

Output at t : $Y_t = \Big(\sum_i (\zeta_i \, A_{t-\tau_i} \, l_{i,t-\tau_i})^{\frac{\sigma-1}{\sigma}}\Big)^{\frac{\sigma}{\sigma-1}}$ labour hired at $t - \tau_i$

Labour at t : $L_t = \sum_i l_{i,t}, \qquad \zeta_i \, A_t \, l_{i,t} = y_{i,t+\tau_i}$ delivered at $t + \tau_i$

Time to build in the data



- Inventory/COGS, 269 industries
- N_i matches each lag
- $\kappa = 0.9018$
- Model: $\tau_i \in \{0, \ldots, 13\}$, mode 2

Expected shocks

Optimal rate path

Planner's problem

$$\max_{\{i_t\}} \; U = \sum_t \beta^t \, \Xi_t \big[u(C_t) - \chi \, v(N_t) \big]$$

$$c_t^e = \varphi_{ya} \, a_t, \qquad \varphi_{ya} = \frac{1+\eta}{\gamma+\eta} = 0.75, \qquad r_t^n = \rho + \gamma \, \Delta y_t^n - \mathsf{dem}_t, \qquad \pi_t = -\Delta \tilde{a}_t$$

- Shocks: TFP -3%, demand -1%
- No welfare cost of inflation

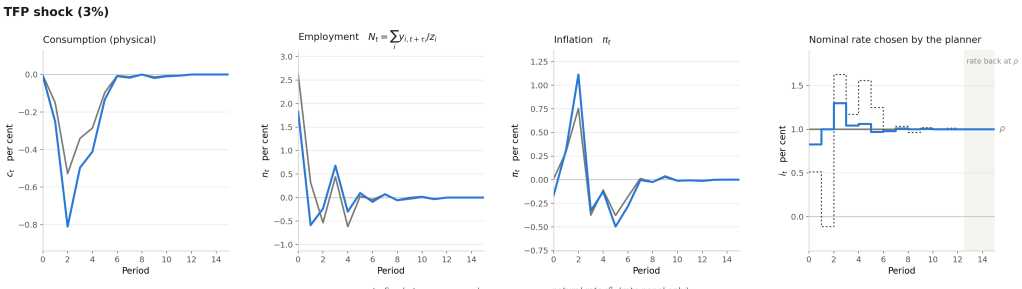
Channel: a rate cut lowers prices for $\max \tau_i$ periods

$i_0 = 0.5\%$ for one period, no shock.

t	0	1	2	3	4	5
$\log P_t$, %	-0.495	-0.437	-0.258	-0.134	-0.045	-0.012
π_t, %	-0.495	0.058	0.179	0.124	0.089	0.033

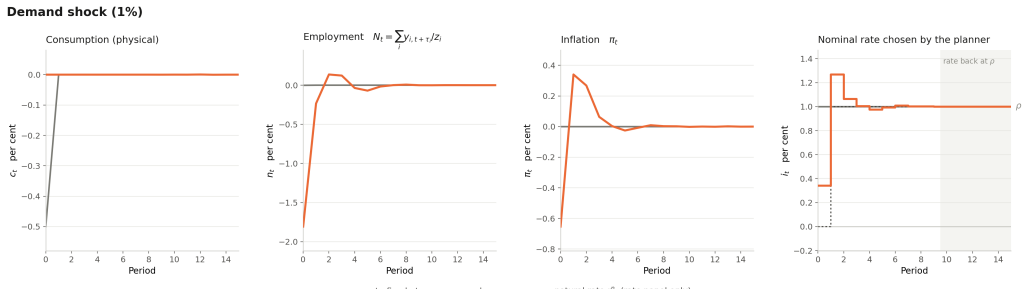
Prices move for **13 periods**. A cut is deflationary.

Expected TFP shock (3%)



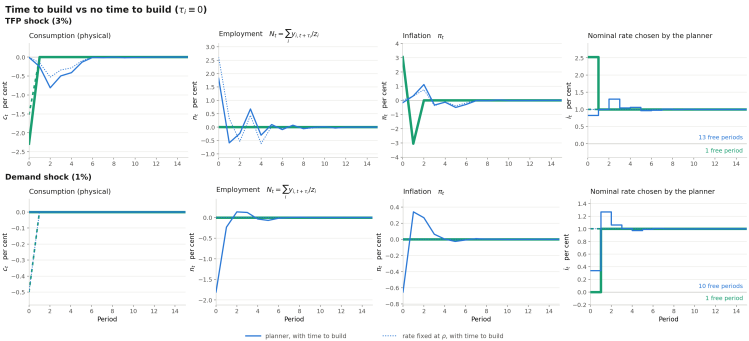
- Peg too loose: c trough -0.53%, efficient -0.81%
- $i = 0.83,\ 1.00,\ \mathbf{1.30},\ 1.04,\ 1.06\,\%$

Expected demand shock (1%)



- $i = \mathbf{0.34},\ \mathbf{1.27},\ 1.06,\ 1.00\,\%$: cut, then overshoot
- Hike undoes the deflation from the cut

Time to build vs none



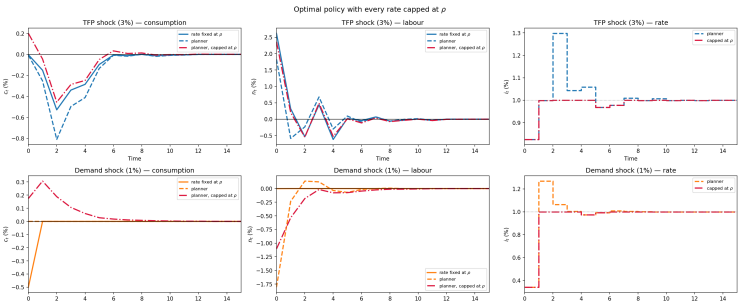
$\tau_i \equiv 0$

one move, $i_t = r_t^n$

Time to build

10–13 periods, oscillating

Cuts only: rate capped at ρ



- Consumption overshoots
- TFP: **worse than the peg**

γ sets the sign of the TFP gap

Peg is loose iff

$$\gamma\varphi_{ya} = \frac{\gamma(1+\eta)}{\gamma+\eta} > 1 \iff \gamma > 1$$

$\gamma = 1$: peg is optimal.

γ	$\gamma\varphi_{ya}$	gap, %
0.50	0.60	-0.82
0.75	0.82	-0.23
1.00	1.00	$+0.03$
1.50	1.29	$+0.22$
2.00	**1.50**	$+0.28$
3.00	1.80	$+0.30$

Unexpected shocks

Quantities committed in advance

Who can respond

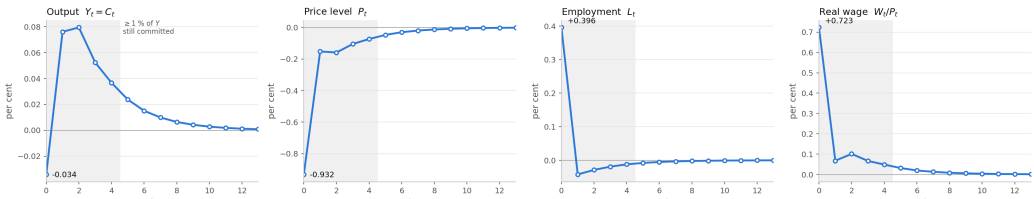
$y_{i,t}$ chosen at $t - \tau_i$:

- $\tau_i > t$: **locked** at y_i^{ss}, $p_{i,t} = P_t \left(y_i^{ss}/Y_t\right)^{-1/\sigma}$
- $\tau_i \leq t$: **free**, prices at cost

Labour is free from $t = 0$.

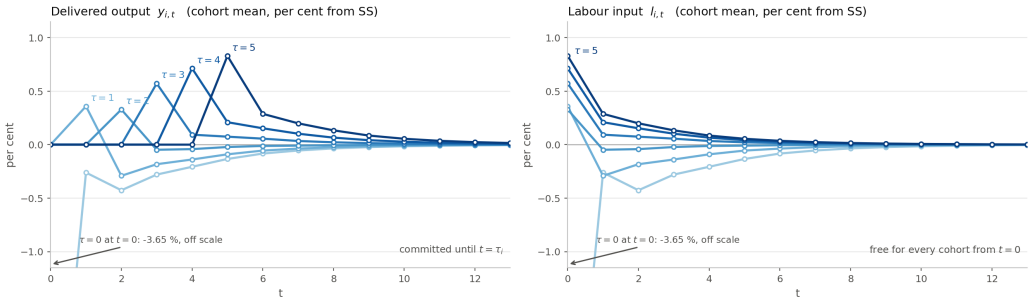
t	free	locked share of Y
0	1	99.1%
1	32	77.2%
2	129	32.5%
13	269	0%

Unexpected demand shock (1%), peg



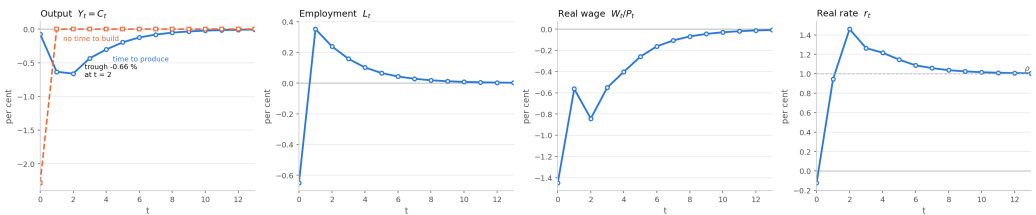
- Impact: $P_0 = -0.93\%$, $Y_0 = -0.03\%$

Unexpected demand shock: by production lag



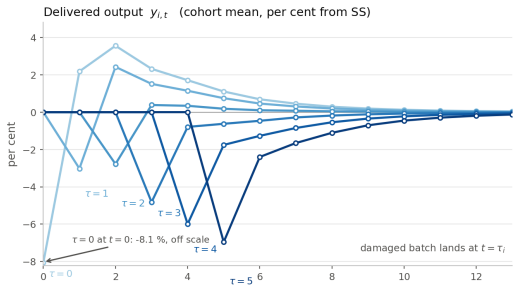
- Deliveries move at $t = \tau_i$; labour moves at $t = 0$

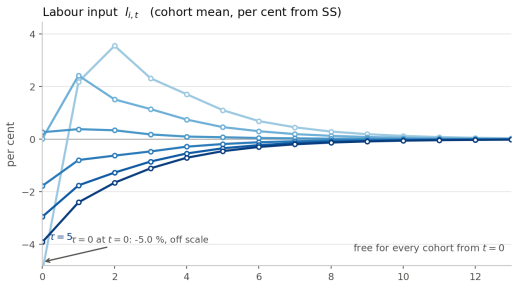
Unexpected TFP shock (3%, one period)



- Trough -0.66% **at** $t = 2$; no time to build: -2.28% **at** $t = 0$

Unexpected TFP shock: by production lag





- Damaged batch arrives at $t = \tau_i$

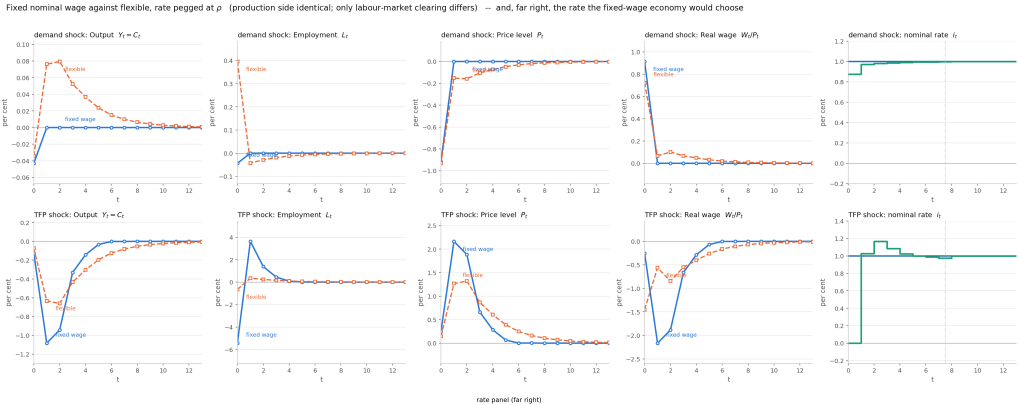
Reading the by-lag figures

One line per production lag $\tau = 0, \ldots, 5$ (mean over firms with that τ).

$$\text{left:} \quad 100 \log \frac{y_{\tau,t}}{y_{\tau}^{ss}} \qquad\qquad \text{right:} \quad 100 \log \frac{l_{\tau,t}}{l_{\tau}^{ss}}$$

- Left: **output delivered at t**, relative to steady state
- Right: **labour hired at t**, relative to steady state
- Line τ: delivered output cannot move before $t = \tau$; labour moves from $t = 0$

Fixed vs flexible wage, peg



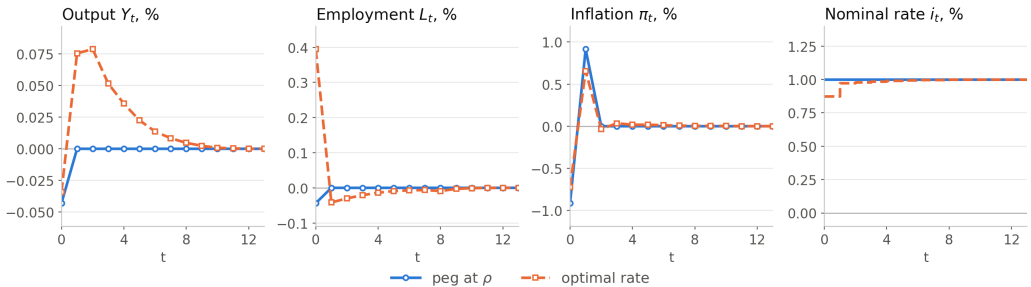
- TFP: peak employment 5.4% vs 0.65%

Optimal rate, fixed wage

	peak $\lvert Y\rvert$, %		peak wedge, %		i_0^*	gain, % of C
	peg	opt.	peg	opt.		
demand	0.043	0.079	1.09	0.02	0.875%	0.00009
TFP	1.083	0.828	10.88	5.14	0%	0.0107

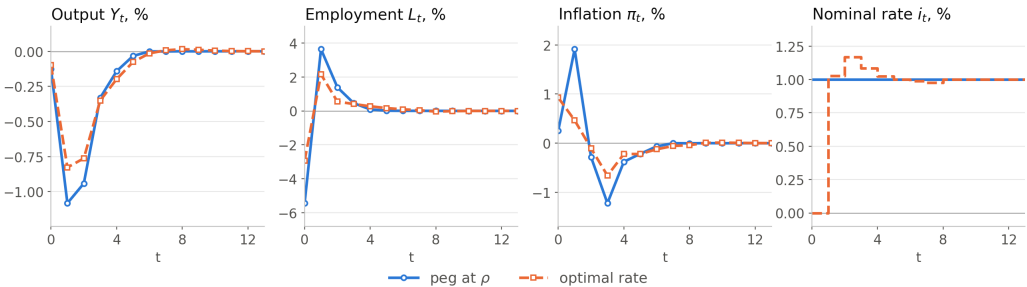
- Demand: small cut closes the wedge
- TFP: **zero bound at $t = 0$**, then above ρ

Optimal rate, fixed wage: demand shock



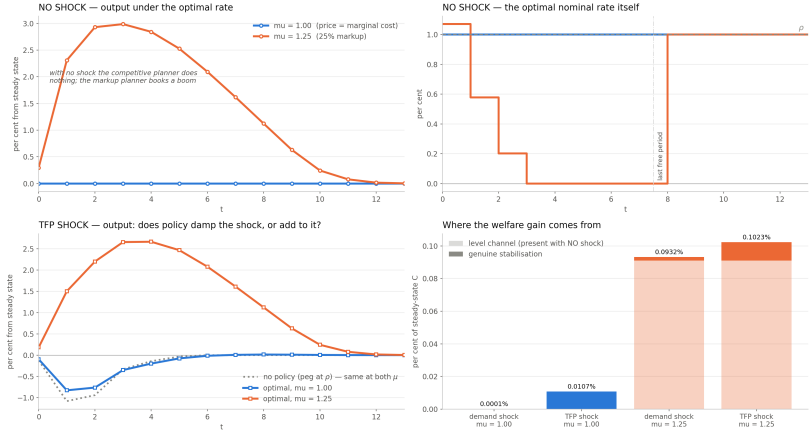
- Cut to 0.875% **at** $t = 0$; wedge $1.09\% \to 0.02\%$

Optimal rate, fixed wage: TFP shock



- **Zero at $t = 0$**, then above ρ; peak employment $5.4\% \to 2.9\%$

Optimal rate, fixed wage: with a markup



- $\mu = 1.25$, no shock: rate to zero, **output** $+3\%$; $\mu = 1$: stays at ρ
- Gain 0.09–0.10% of C, about 90% of it without any shock

Markup: the boom comes from χ

$$\chi N^{\eta} C^{\gamma} = \frac{W}{P} = \frac{A}{\mu} \qquad \Longrightarrow \qquad C = N = A = 1 \text{ needs } \chi = 1/\mu$$

no shock, $\mu = 1.25$, 8 free periods	$\chi = 1$	$\chi = 1/\mu$
expected shocks	$i^* = \rho$	$i^* = 0$, C $+4.5\%$
unexpected shocks	$i^* = \rho$	$i^* \to 0$, Y $+3.0\%$

- $\chi = 1/\mu$: employment below efficient, planner cuts
- $\chi = 1$: efficient steady state (subsidy offsets the markup); μ drops out
- Normalising the steady state alone changes nothing

$\chi = 1/\mu$: the boom lasts as long as the window

No shock, unexpected-shock model, 8 $\to$ 16 free periods.

μ	i_0^*, %	periods at zero	peak Y, %
1.00	$1.00 \to 1.00$	$0 \to 0$	$0 \to 0$
1.05	$1.07 \to 1.17$	$0 \to 0$	$0.82 \to 1.13$
1.15	$1.18 \to 1.48$	$2 \to 3$	$2.26 \to 3.21$
1.25	$1.07 \to 1.72$	$5 \to 7$	$2.99 \to 4.86$
1.50	$0.41 \to 1.65$	$7 \to 13$	$3.51 \to 6.76$

- Wage fixed forever, inflation free: nothing ends the boom but the peg

$\chi = 1/\mu$: the response to the shock

i_0^*(shock) $-$ i_0^*(no shock), pp, 8 $\to$ 16 free periods.

μ	demand	TFP
1.00	$-0.125 \to -0.125$	$-1.00 \to -1.00$
1.15	$-0.128 \to -0.129$	$-1.18 \to -1.48$
1.25	$-0.132 \to -0.131$	$-1.07 \to -1.72$
1.50	$-0.144 \to -0.137$	$-0.41 \to -1.65$

- Demand: same cut at every μ and window
- TFP: $i_0^* = 0$ at every μ
- Gain at $\mu = 1.25$, % of C: boom 0.091; shock 0.002 (demand), 0.011 (TFP)

Inflation bias

- $\chi = 1/\mu$: steady state inefficient, planner cuts whenever it is free
- Return to ρ is imposed: longer window, longer boom
- **Classical inflation-bias problem**
- Next: solve the commitment problem in this setting

Summary

- **Expected**: rate moves echo for 13 periods; optimal path cuts, then overshoots
- Planner reaches $|c - c^e| \sim 10^{-6}$; gain $\sim 10^{-4}\%$ of C
- **Unexpected** demand: prices -0.93%, output -0.03%
- **Unexpected** TFP: trough at $t = 2$, not on impact
- Rate neutral with flexible wage, not with fixed wage